

Lecture 2

VLSI Design Flow overview

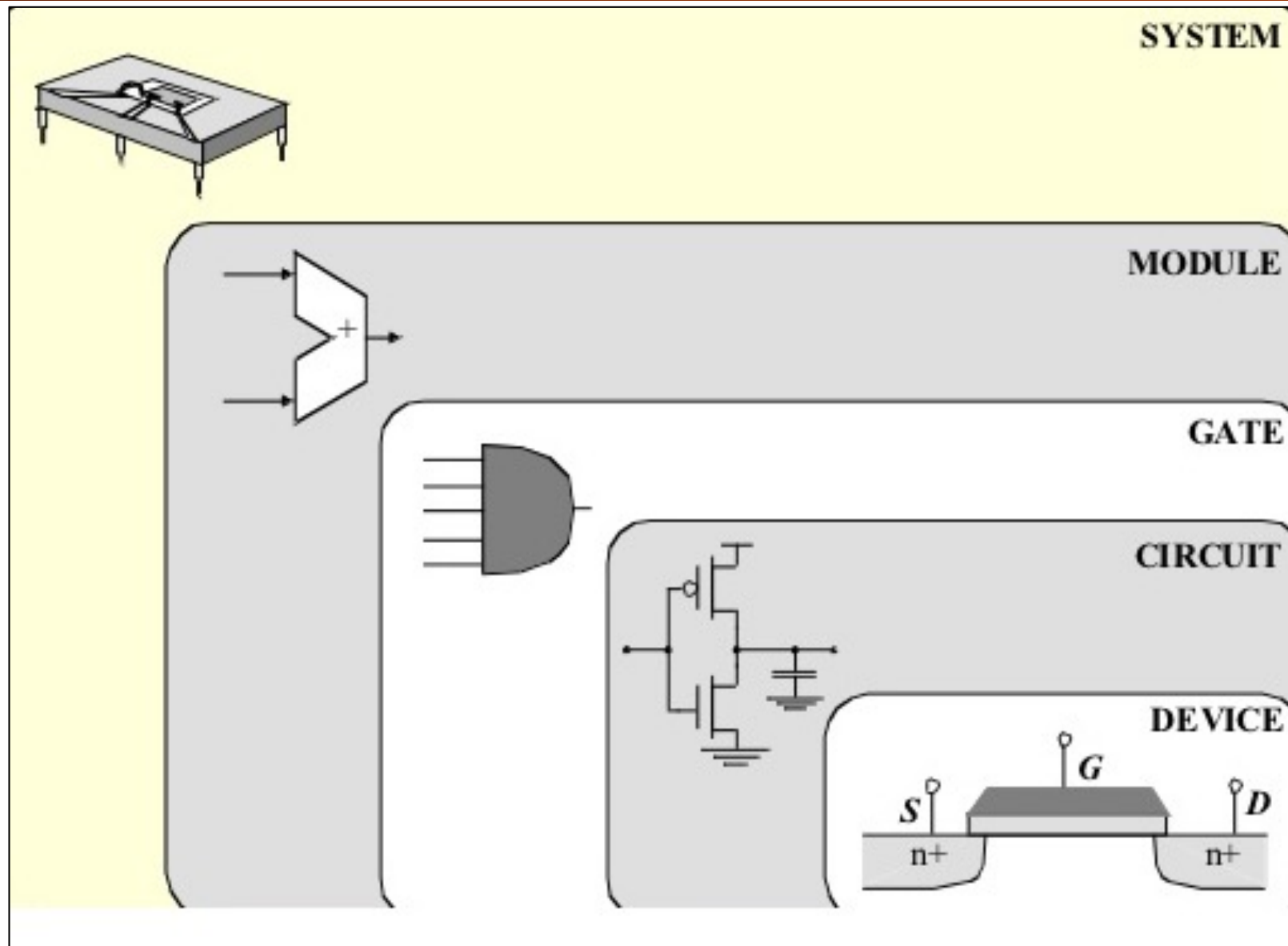
Prof Peter YK Cheung

Department of Electrical & Electronic Engineering

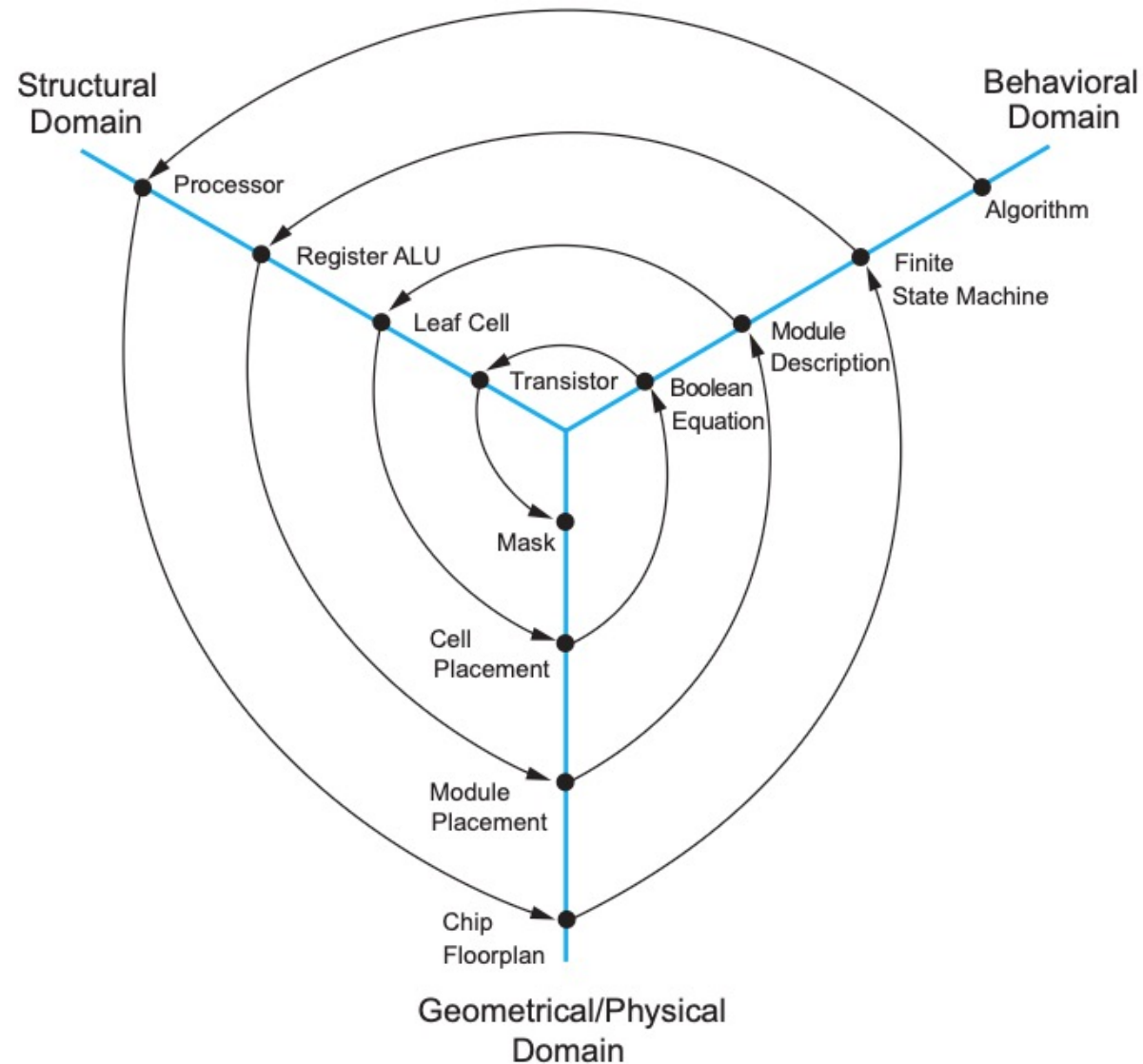
URL: www.ee.ic.ac.uk/pcheung/teaching/EE4-VLSI/

E-mail: p.cheung@imperial.ac.uk

Levels of Abstraction in Digital Design

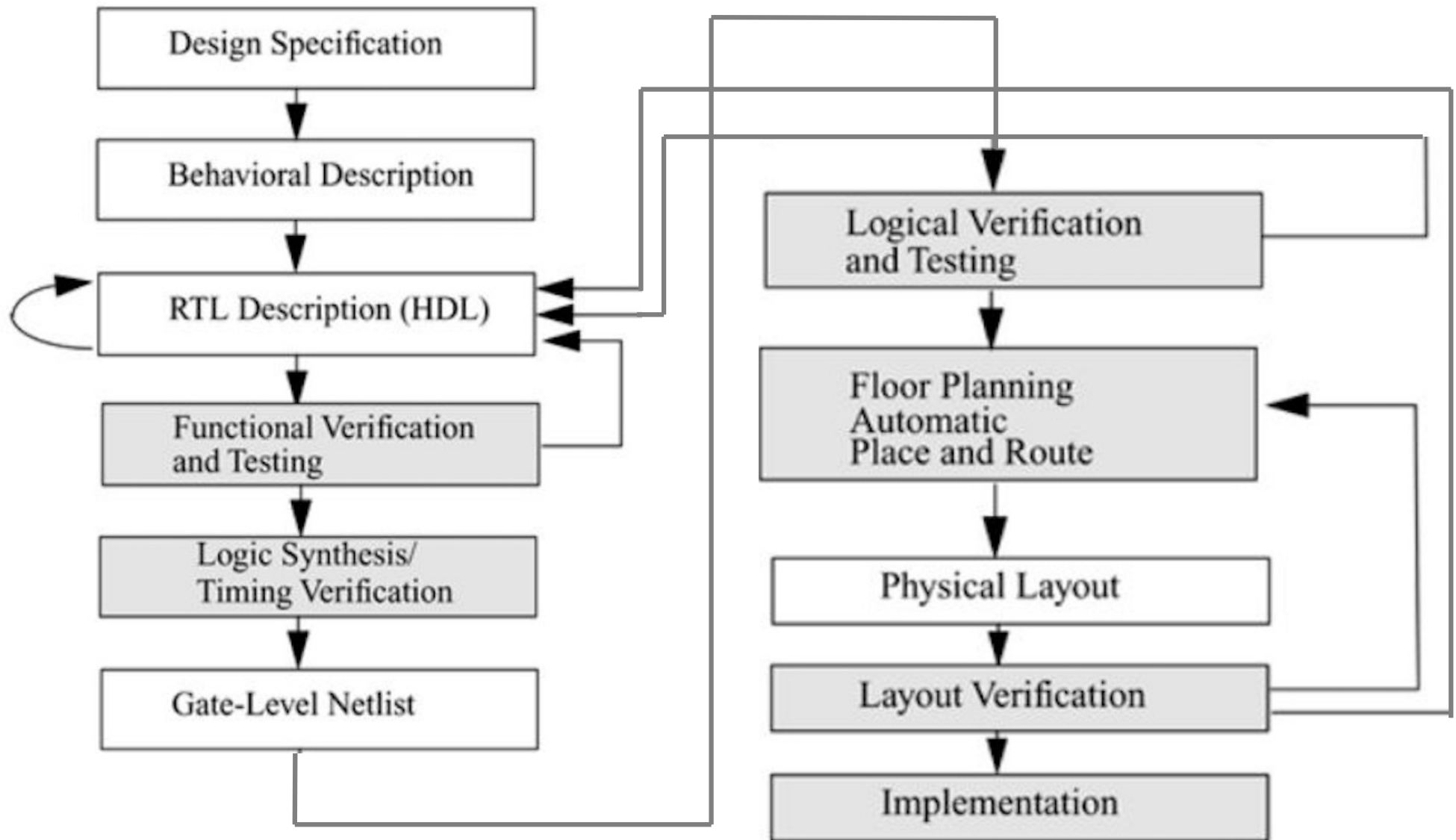


Levels of Abstraction in Digital Design

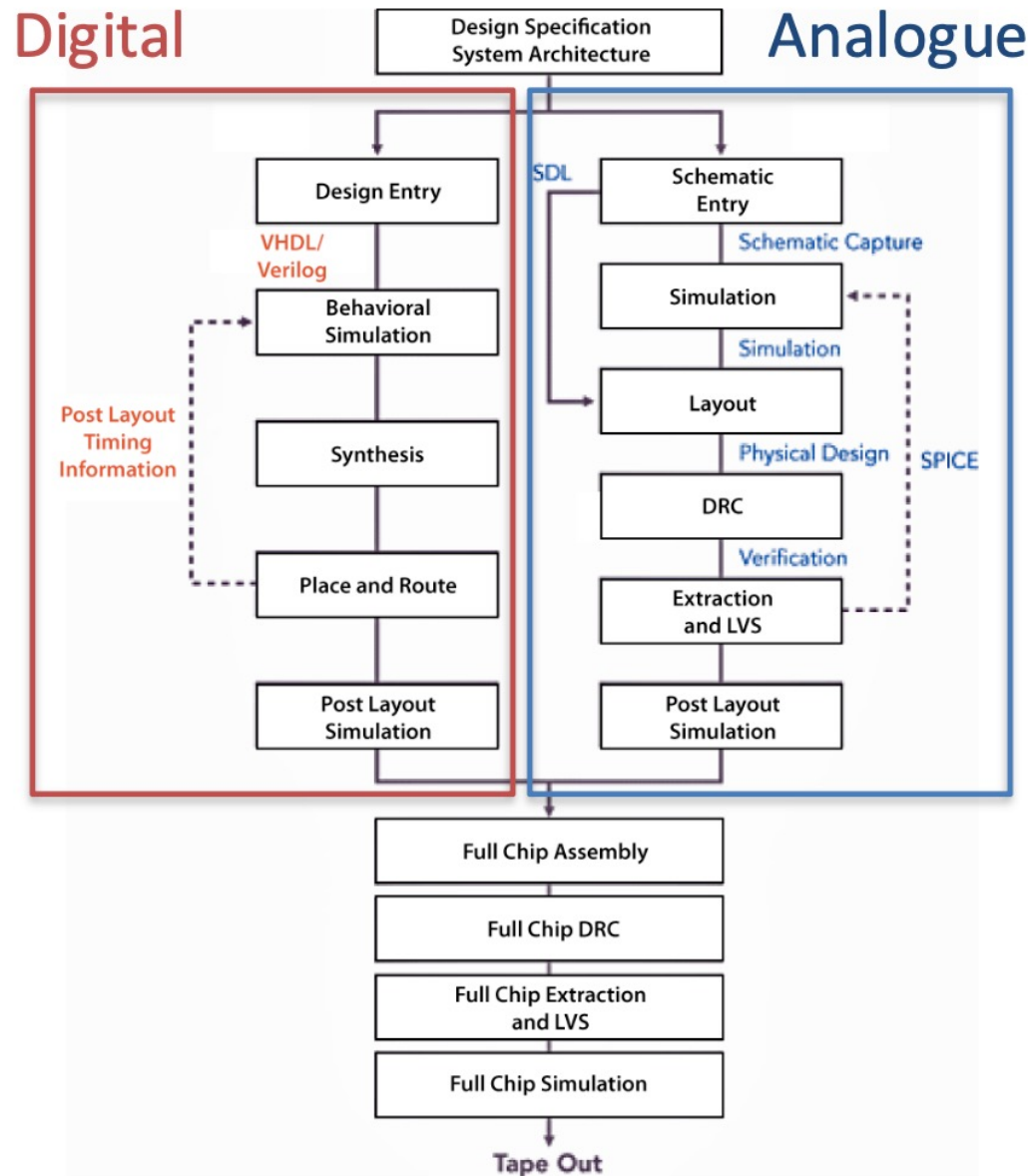


Source: Gajski 1983 & Kang 2003

Full-Custom VLSI Design Flow

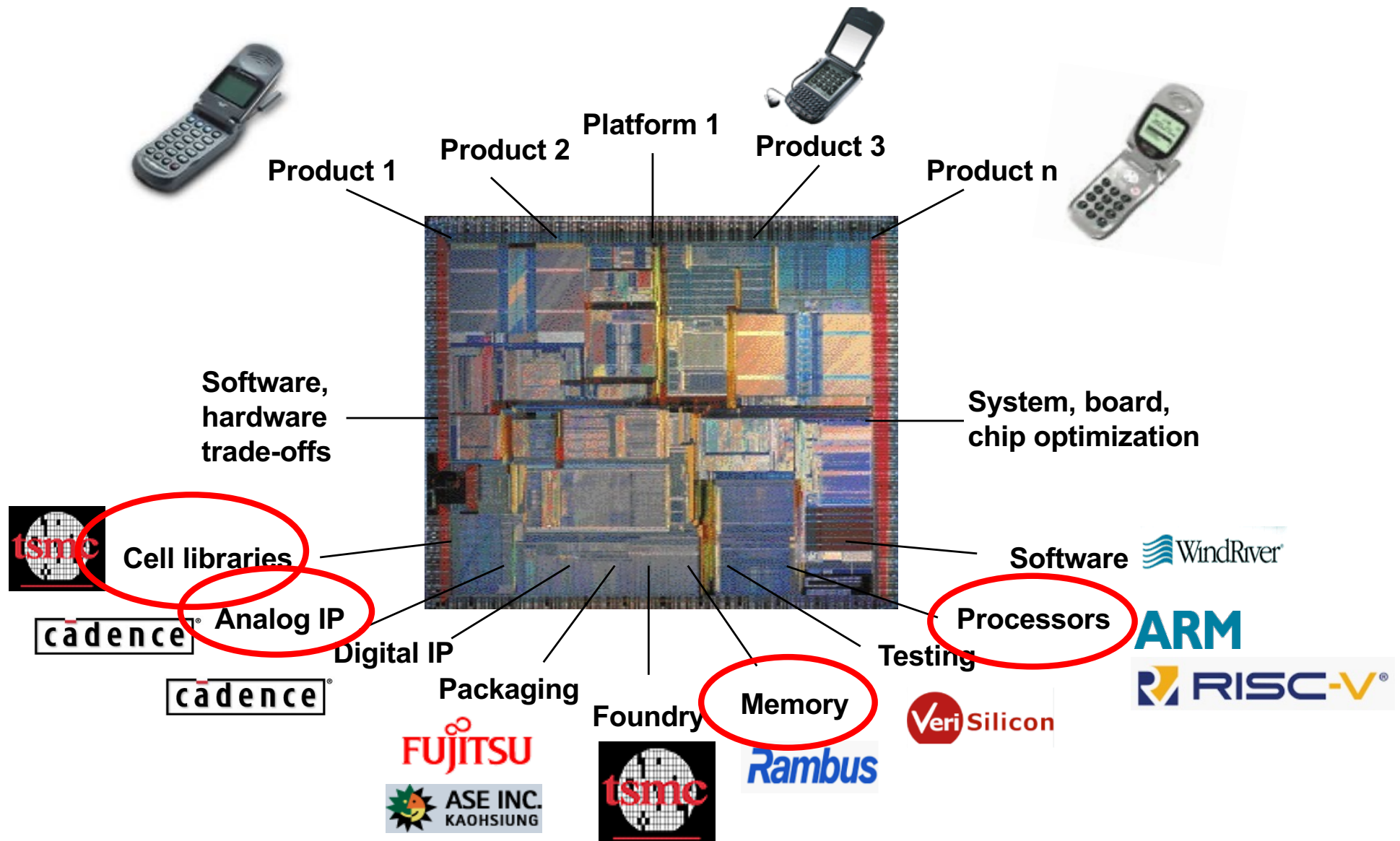


Mixed signal VLSI Design Flow



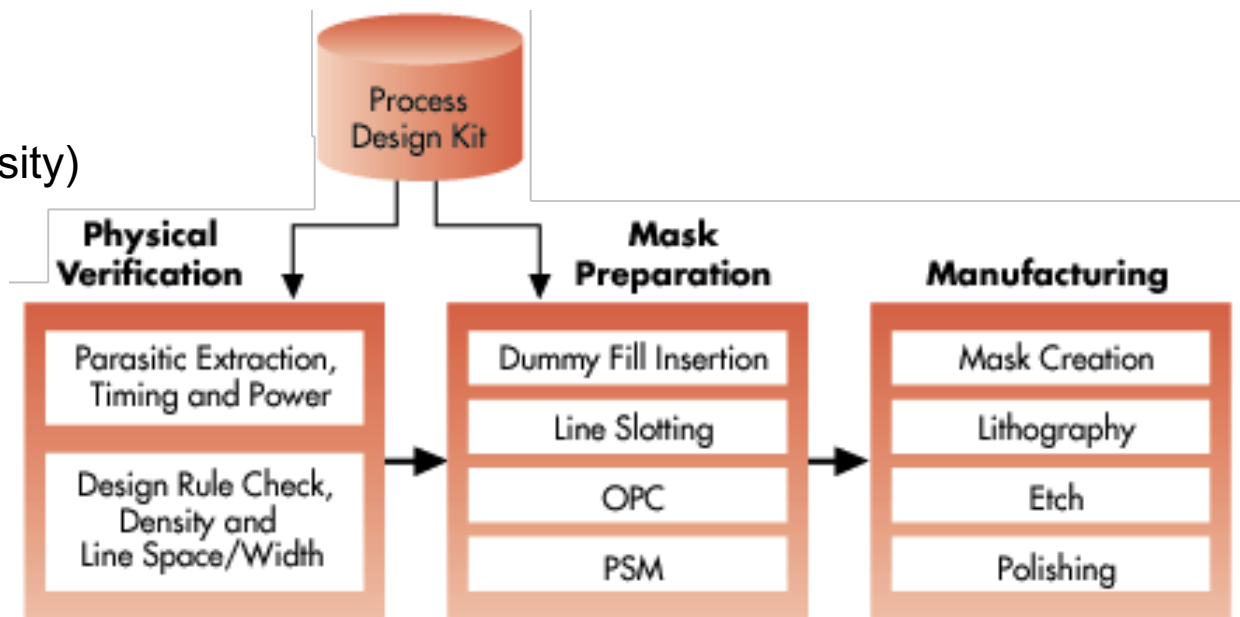
Source: Notes from T Constandinou

IP based VLSI Design

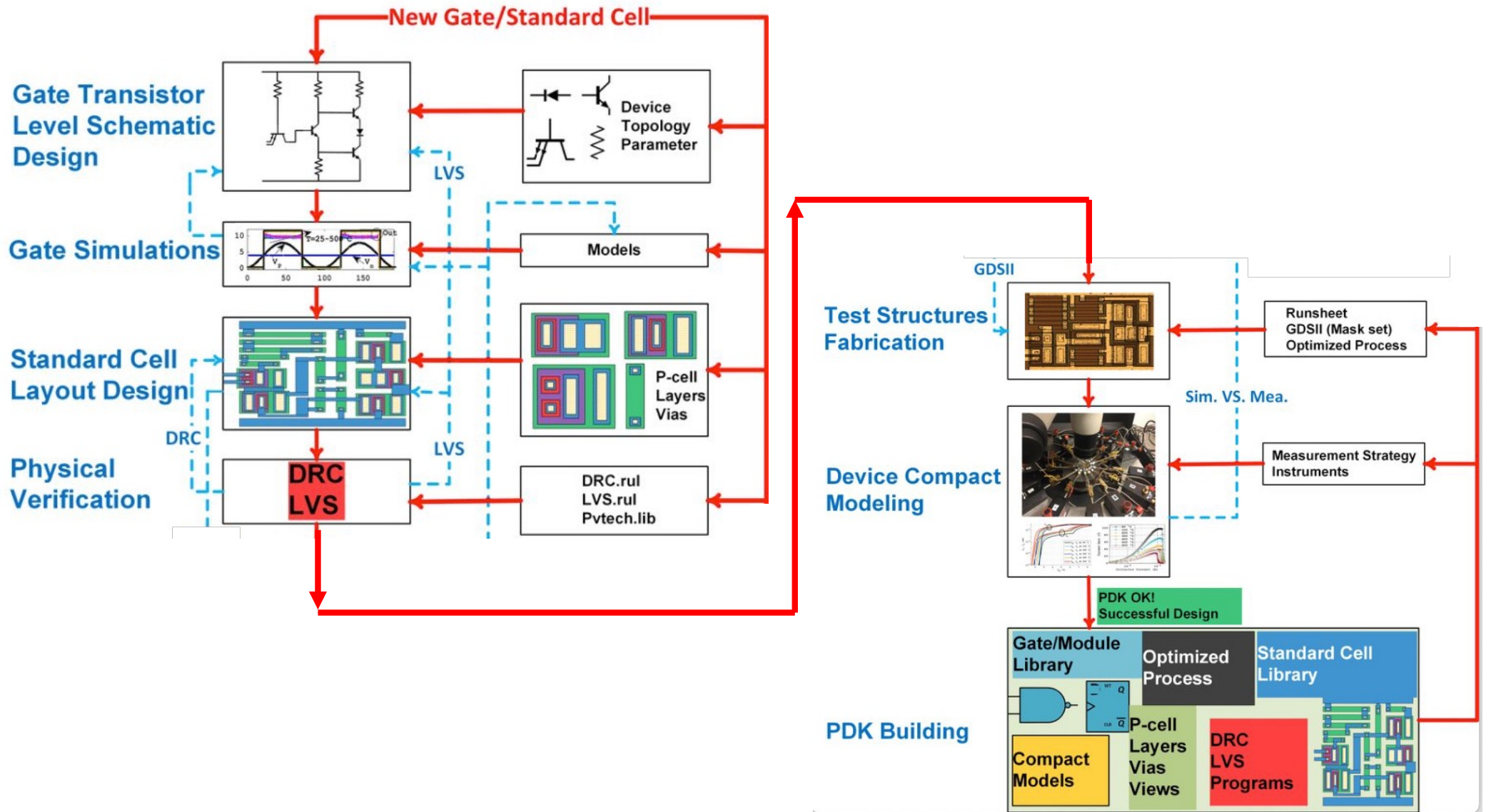


What is a Process Design Kit (PDK)

- ◆ Link between chip maker (foundry) and EDA tool supplier (TSMC → Synopsys)
- ◆ PDK is for a specific process (we use TSMC 65nm low power process)
- ◆ PDK consists for set of files in standardised format for different design steps:
 - Primitive devices (e.g. transistors, capacitor etc.)
 - Device parameters
 - Device symbol
 - Technology related information
 - Layers and layer names
 - Appearance (colour)
 - Constraints (e.g. current density)
 - Verification information
 - Design rule checking (DRC)
 - Layout vs Schematic (LVS)
 - Device extraction
 - Simulation models
 - Standard cell libraries
 - Documentation



PDK Build Process



Source: Gupta

Synopsys Fusion Compiler

COMMON & INTERLEAVED ENGINES

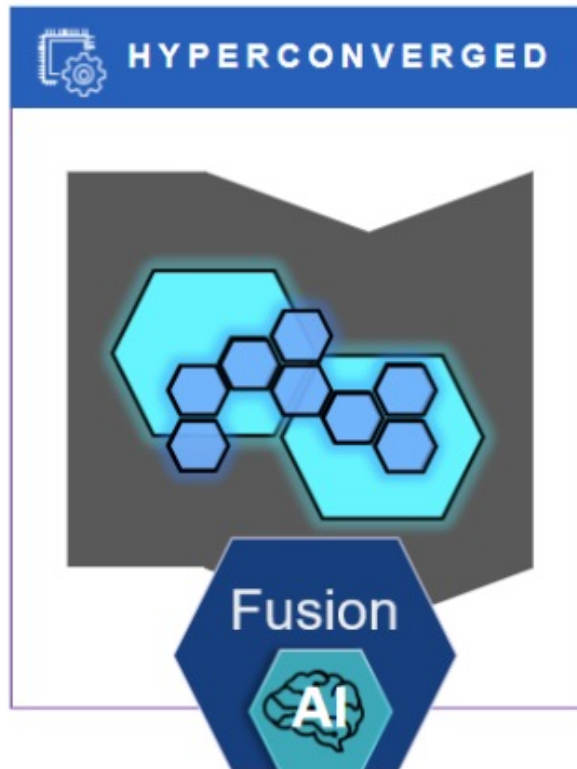
LOGIC SYNTHESIS

PLACEMENT

CLOCK

ROUTE

AUTONOMOUS OPT.



NATIVE GOLDEN SIGNOFF BACKBONE

STARRC EXTRACTION

PRIMETIME ANALYSIS

PRIMEPOWER ANALYSIS

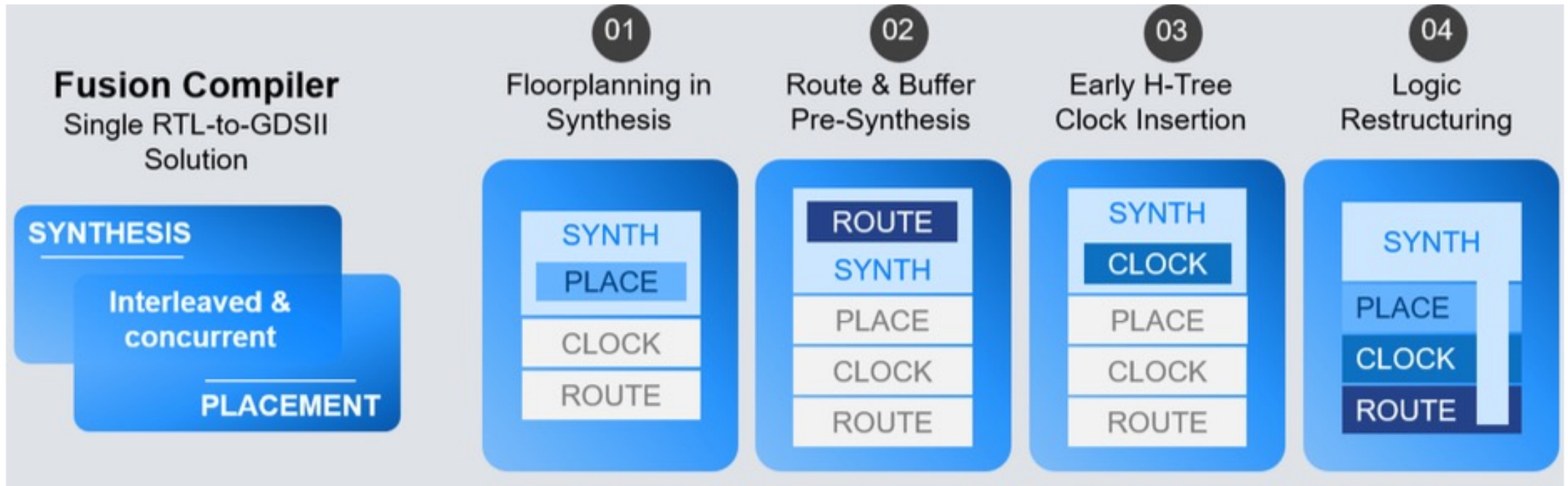
REDHAWK ANALYSIS

ANALYTICAL DECISIONS

- ◆ Unified data model operating in a coherent environment.
- ◆ Avoid translation error.
- ◆ Constraints are consistent throughout.
- ◆ Speed up the learning curve of engineers.

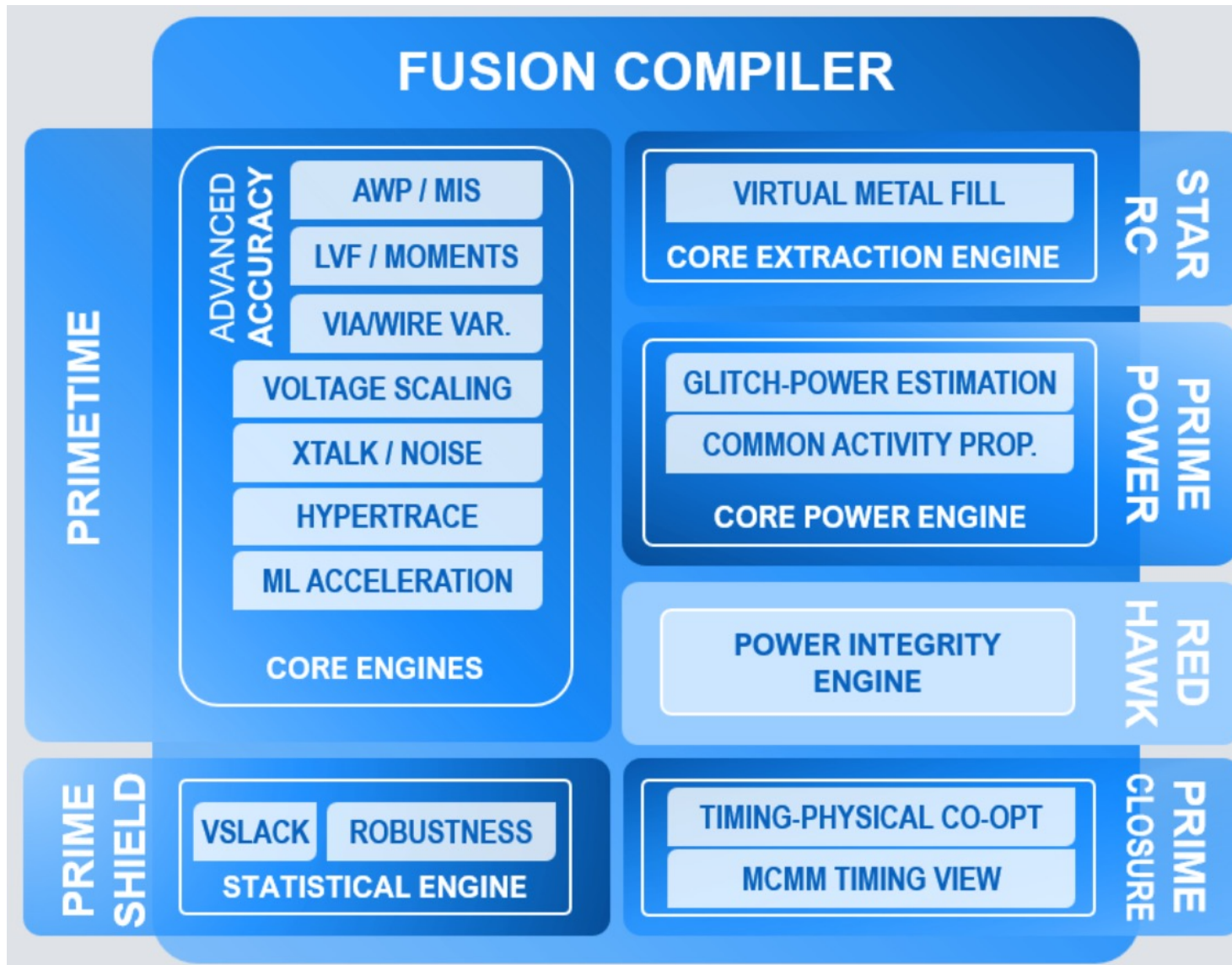
source: Fusion Compiler User Guide

Fusion Compiler Shell Engines



source: Fusion Compiler User Guide

Power Performance Area Optimization

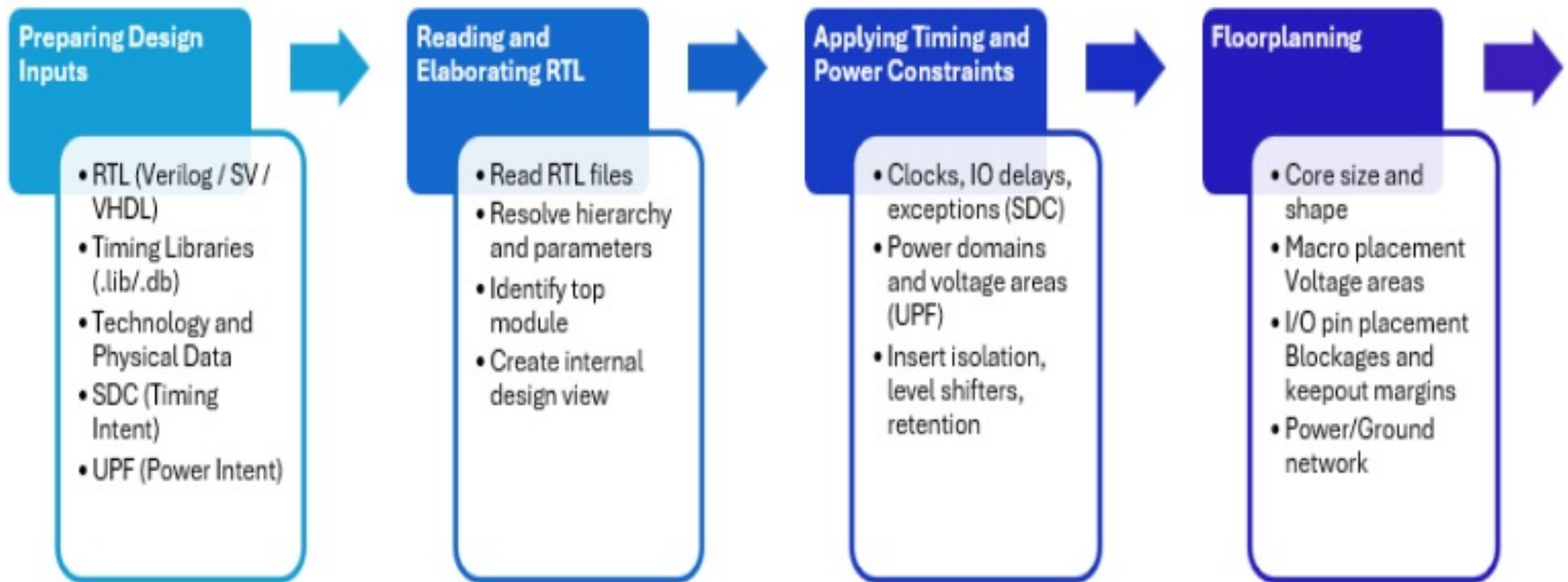


Trade off between:

- ◆ Power
- ◆ Performance
- ◆ Area
- ◆ FC iterates between these to resolve conflicts

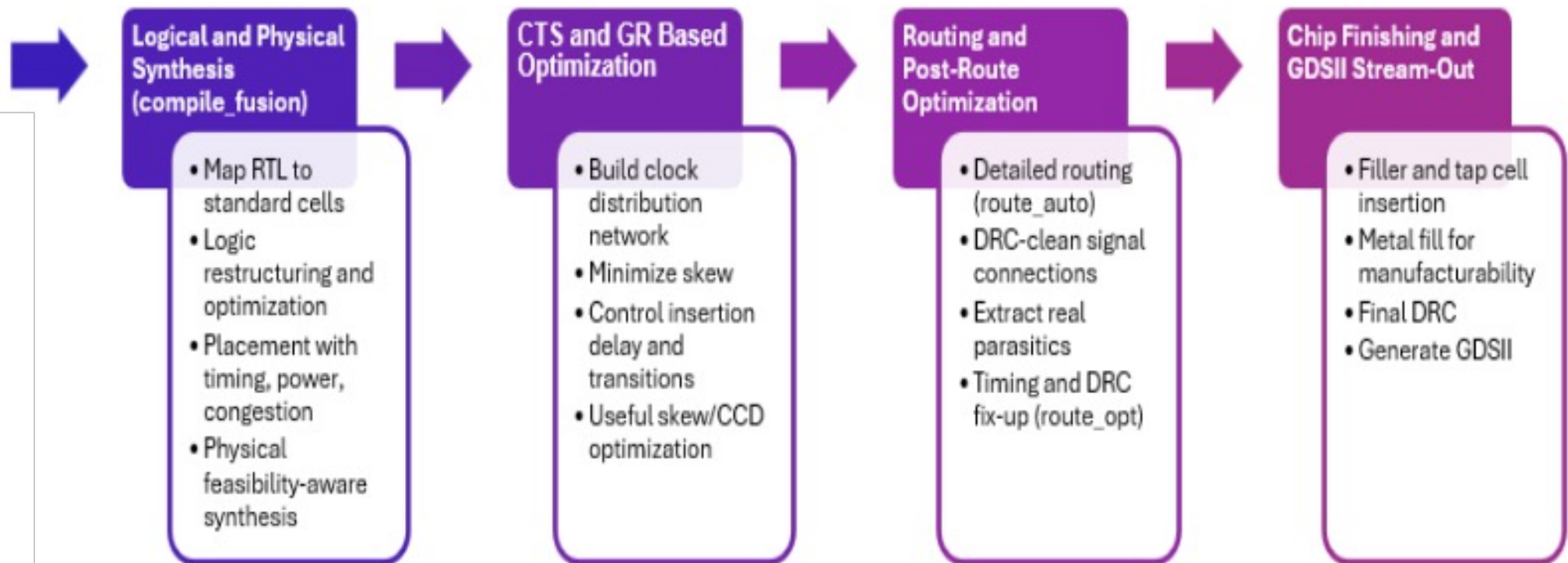
source: Fusion Compiler User Guide

RTL to GDSII Design Flow in FC (1 of 2)



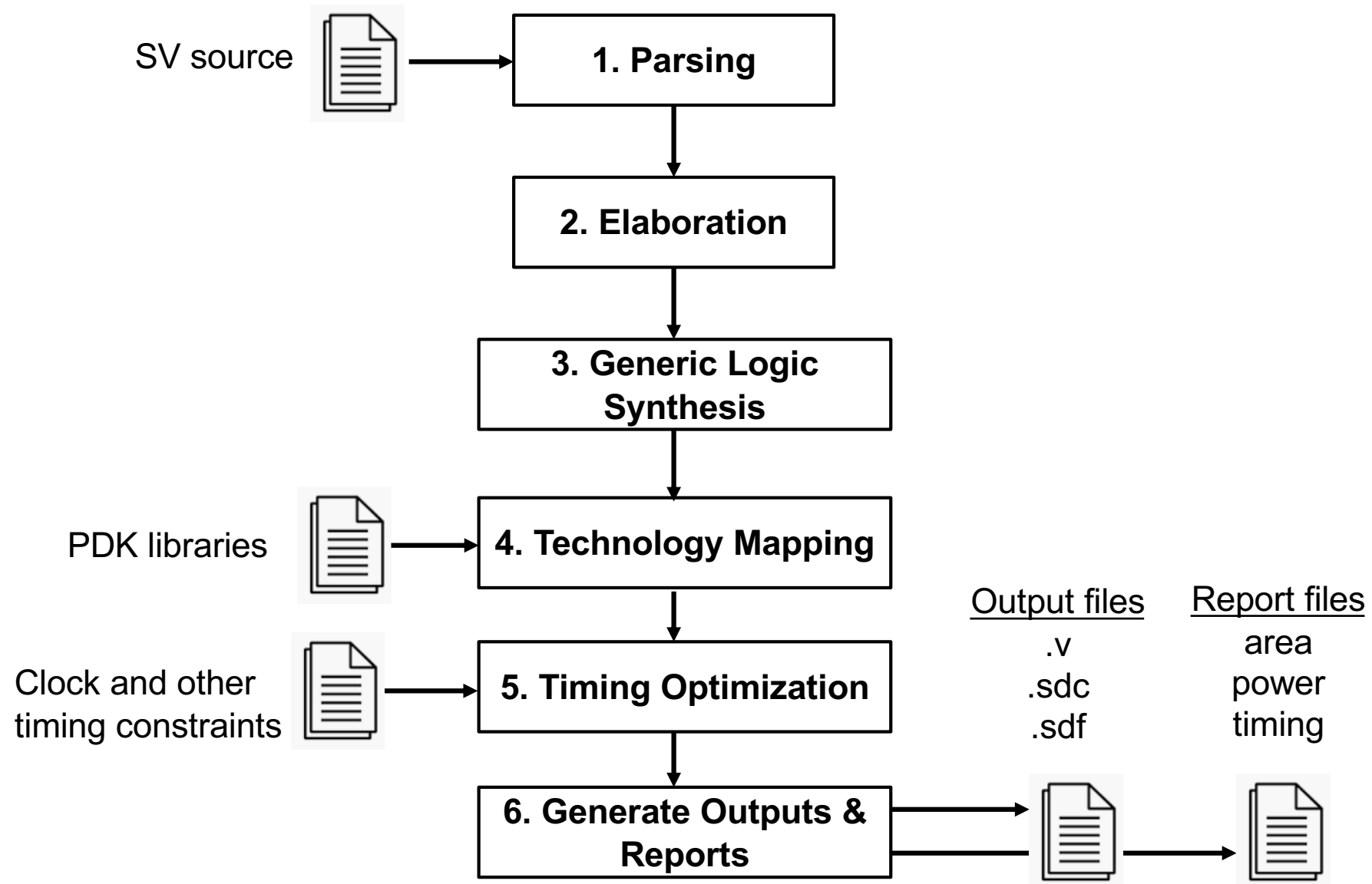
source: Fusion Compiler User Guide

RTL to GDSII Design Flow in FC (2 of 2)



source: Fusion Compiler User Guide

Lab 1 > Task 2: RTL to Standard Cell Synthesis



Synthesize LFSR4

```
module lfsr4 (  
    input logic      clk,      // clock  
    input logic      rst,      // reset  
    output logic [3:0] data_out // pseud  
);  
  
logic [4:1] sreg;  
  
always_ff @ (posedge clk, posedge rst)  
    if (rst)  
        sreg <= 4'b1;  
    else
```

RTL source .sv

```
module lfsr4(clk, rst, data_out);  
    .....  
    wire n_0, n_1;  
    DFSNQD1BWP7T \sreg_reg[1] (.SDN (n_0), .CP (clk  
        | (data_out[0]));  
    CKXOR2D0BWP7T g16__8780(.A1 (data_out[2]), .A2  
        | (n_1));  
    DFCNQD1BWP7T \sreg_reg[4] (.CDN (n_0), .CP (clk  
        | .Q (data_out[3]));  
    DFCNQD1BWP7T \sreg_reg[3] (.CDN (n_0), .CP (clk  
        | .Q (data_out[2]));  
    DFCNQD1BWP7T \sreg_reg[2] (.CDN (n_0), .CP (clk  
        | .Q (data_out[1]));  
    INVND0BWP7T g20(.I (rst), .ZN (n_0));  
endmodule
```

Synthesized circuit .v

Fusion Compiler
syn_logical.tcl

```
#####  
## Design  
#####  
  
set DESIGN    lfsr4  
set CLK_PORT  clk  
  
set RTL_FILES src/$DESIGN.sv  
set SDC_FILE  constraints/$DESIGN.sdc  
  
#####  
## Settings you may want to change  
#####  
# The clock is not here. It lives in t  
  
set CORE_UTIL    0.6      ;# fractio  
set ASPECT       1.0      ;# core he  
set CORE_OFFSET  5        ;# core to  
  
Tcl commands/script
```

```
-----  
data required time          0.78  
data arrival time          -0.00  
-----  
slack (MET)                 0.78
```

Timing report

Synthesis Steps (*setup.tcl*) (1)

```
#####
## Settings you may want to change
#####
## Design
#####
# The clock is not here. It lives i

set DESIGN    lfsr4
set CLK_PORT  clk

set RTL_FILES src/$DESIGN
set SDC_FILE  constrain

#####
set CORE_UTIL    0.6
set ASPECT       1.0
set CORE_OFFSET  5

set RING_WIDTH    1.0
set RING_SPACING  0.5
# Centres the ring set
# (CORE_OFFSET - (2 *
set RING_OFFSET  1.25

# Power rails on the 1
set RAIL_LAYER M1

# The cell rails are o
# rather than a crossi
set RING_H_LAYER M3
set RING_V_LAYER M2

#####
## Directories and files
#####

set WORK_DIR work
set OUT_DIR  outputs/$FLOW
set RPT_DIR  reports/$FLOW

set SYN_LIB $WORK_DIR/${DESIGN}_syn.dlib
set PNR_LIB $WORK_DIR/${DESIGN}_${FLOW}.dlib

set SYNTH_V    $OUT_DIR/${DESIGN}_synth.v
set SYNTH_SDC  $OUT_DIR/${DESIGN}_synth.sdc
set LAYOUT_V   $OUT_DIR/${DESIGN}_layout.v
```

Synthesis Steps (*syn_logical.tcl*) (2)

◆ Key inputs required for logic synthesis:

1. RTL source code (.sv)
2. Timing libraries (.lib or .db)
3. Technology files (.tf), parasitic models (.tlup) and physical libraries (.ndm)
4. Timing constraints (.sdc)
5. Power constraints (.upf) – not used in Lab 1

◆ The key steps in performing logic synthesis are:

1. Read and parse RTL
2. Elaborate the RTL design, check all hierarchy, link to library cells and submodules
3. Apply timing (and power) constraints
4. Perform logic synthesis (command: `compile_logical`)
5. Write report and export the design

Output of logic synthesis step

```
module lfsr4 (
    input logic      clk,      // clock
    input logic      rst,      // reset
    output logic [3:0] data_out // pseud
);

logic [4:1]      sreg;

always_ff @ (posedge clk, posedge rst)
    if (rst)
        sreg <= 4'b1;
    else
        sreg <= {sreg[3:1], sreg[3] ^ sreg[4]};

assign data_out = sreg;
endmodule
```

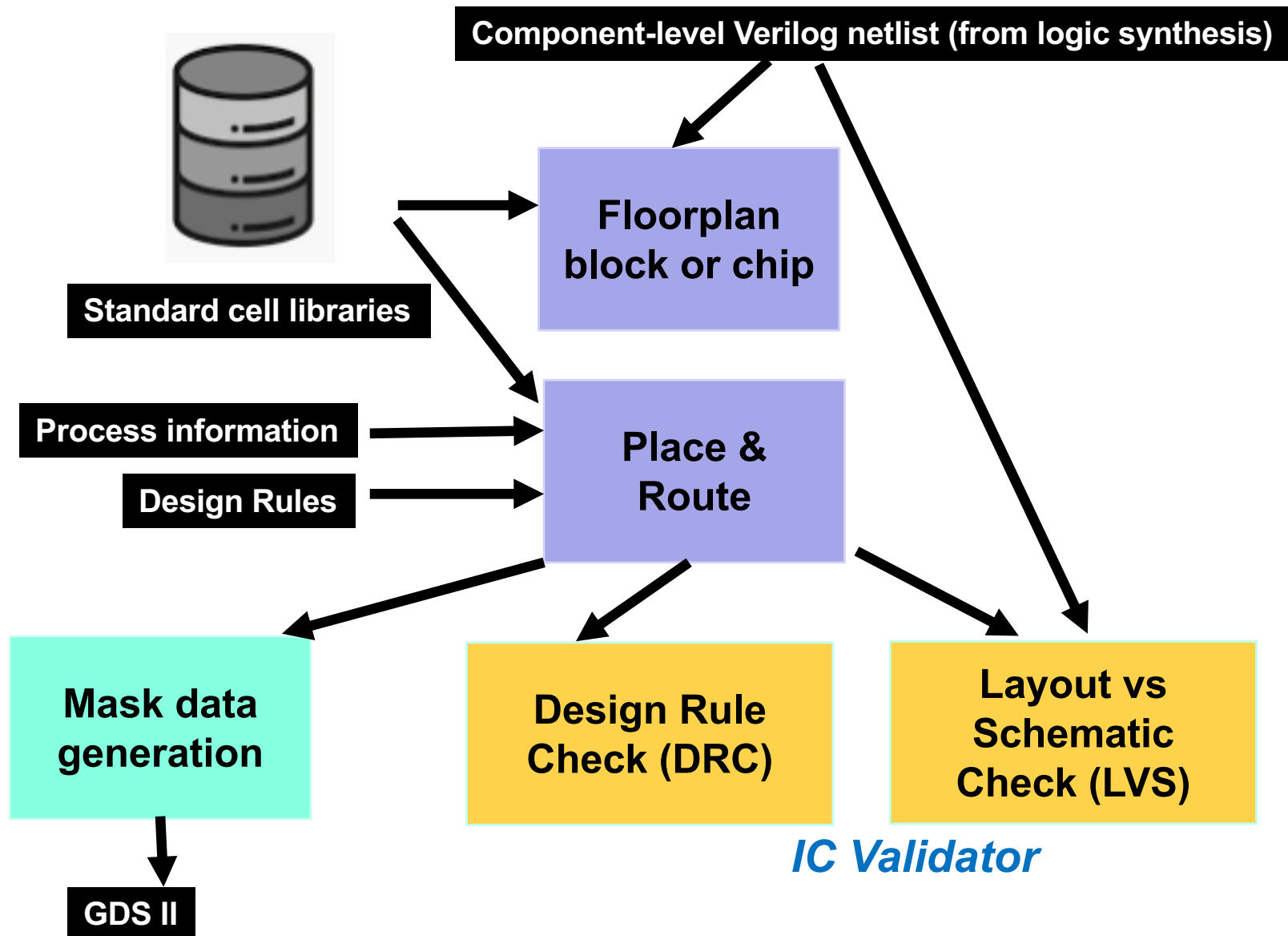
**RTL source
In SystemVerilog**

```
module lfsr4 ( clk , rst , data_out ) ;
input  clk ;
input  rst ;
output [3:0] data_out ;

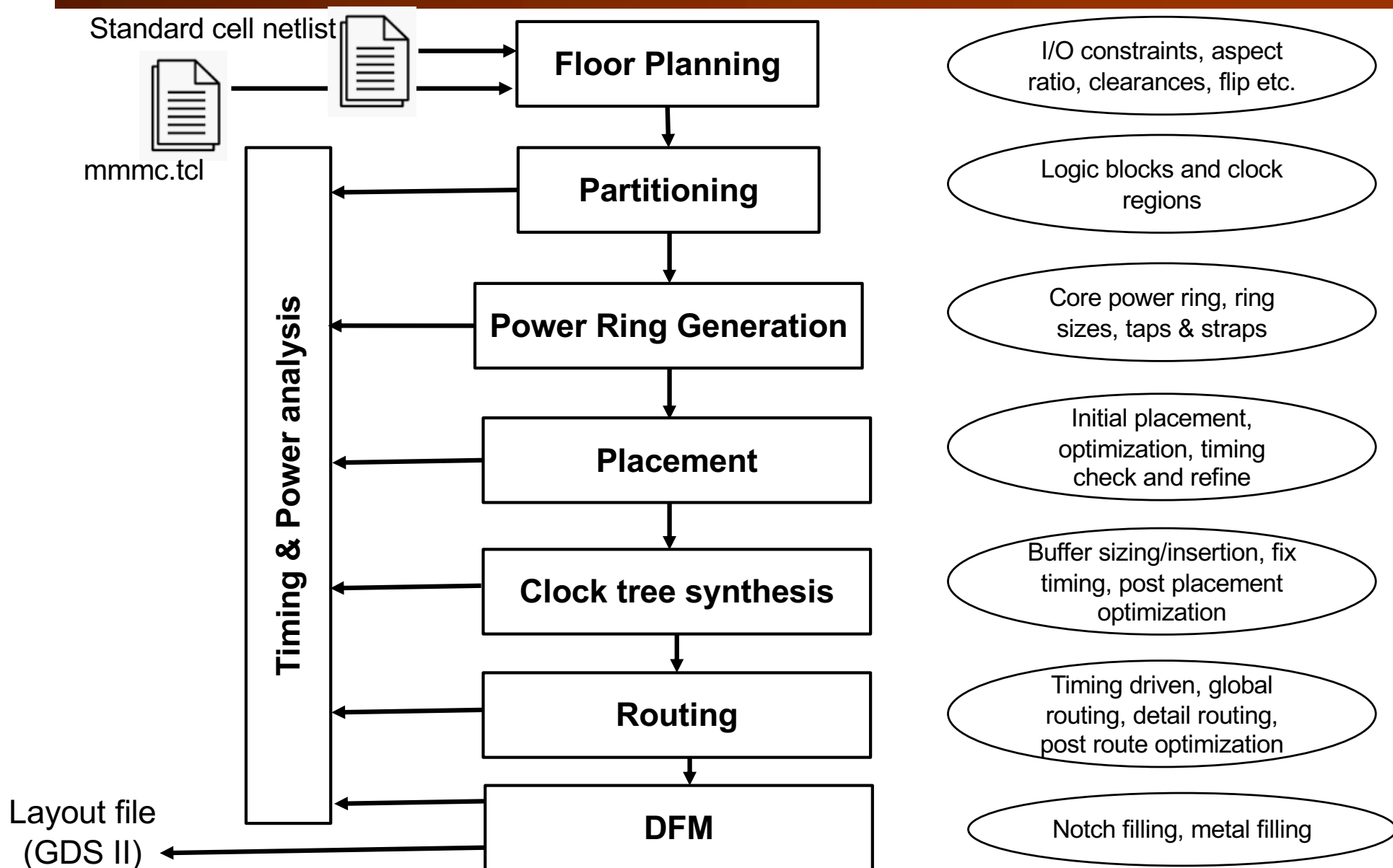
\*SEQGEN.1839103_120_2 \sreg_reg[4] ( .clear ( rst ) , .preset ( 1'b0 ) ,
    .next_state ( data_out[2] ) , .clocked_on ( clk ) , .data_in ( 1'b0 ) ,
    .enable ( 1'b0 ) , .Q ( data_out[3] ) , .synch_clear ( 1'b0 ) ,
    .synch_preset ( 1'b0 ) , .synch_toggle ( 1'b0 ) , .synch_enable ( 1'b1 ) ,
    .scan_enable ( 1'b0 ) , .scan_in ( 1'b0 ) ) ;
\*SEQGEN.1839103_120_2 \sreg_reg[3] ( .clear ( rst ) , .preset ( 1'b0 ) ,
    .next_state ( data_out[1] ) , .clocked_on ( clk ) , .data_in ( 1'b0 ) ,
    .enable ( 1'b0 ) , .Q ( data_out[2] ) , .synch_clear ( 1'b0 ) ,
    .synch_preset ( 1'b0 ) , .synch_toggle ( 1'b0 ) , .synch_enable ( 1'b1 ) ,
    .scan_enable ( 1'b0 ) , .scan_in ( 1'b0 ) ) ;
\*SEQGEN.1839103_120_2 \sreg_reg[2] ( .clear ( rst ) , .preset ( 1'b0 ) ,
    .next_state ( data_out[0] ) , .clocked_on ( clk ) , .data_in ( 1'b0 ) ,
    .enable ( 1'b0 ) , .Q ( data_out[1] ) , .synch_clear ( 1'b0 ) ,
    .synch_preset ( 1'b0 ) , .synch_toggle ( 1'b0 ) , .synch_enable ( 1'b1 ) ,
    .scan_enable ( 1'b0 ) , .scan_in ( 1'b0 ) ) ;
\*SEQGEN.1839103_210_2 \sreg_reg[1] ( .clear ( 1'b0 ) , .preset ( rst ) ,
    .next_state ( N0 ) , .clocked_on ( clk ) , .data_in ( 1'b0 ) ,
    .enable ( 1'b0 ) , .Q ( data_out[0] ) , .synch_clear ( 1'b0 ) ,
    .synch_preset ( 1'b0 ) , .synch_toggle ( 1'b0 ) , .synch_enable ( 1'b1 ) ,
    .scan_enable ( 1'b0 ) , .scan_in ( 1'b0 ) ) ;
WVGTECH_XOR2 C20 ( .I0 ( data_out[2] ) , .I1 ( data_out[3] ) , .Z ( N0 ) ) ;
endmodule
```

**synthesized netlist
in Verilog (before tech map)**

FC Placement and Routing in context

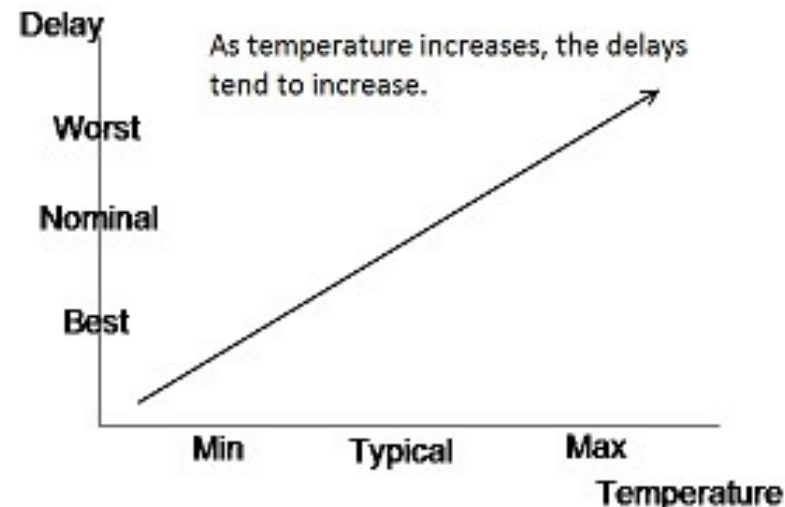
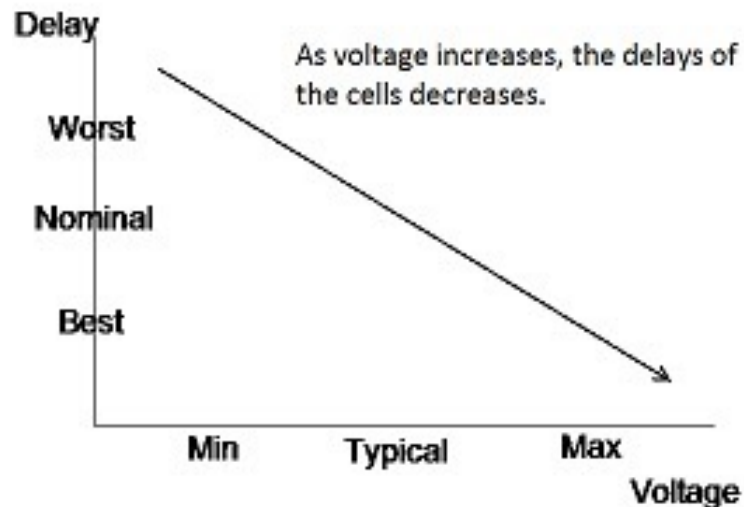
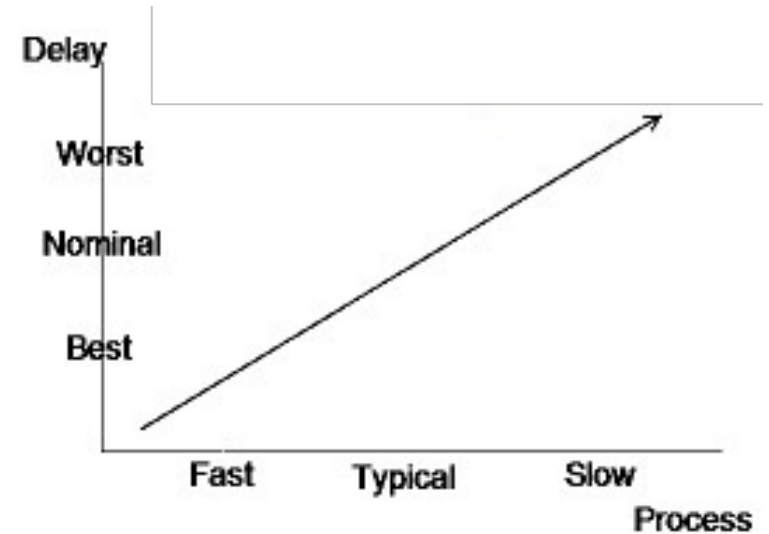


Netlist to Layout Flow



What is PVT?

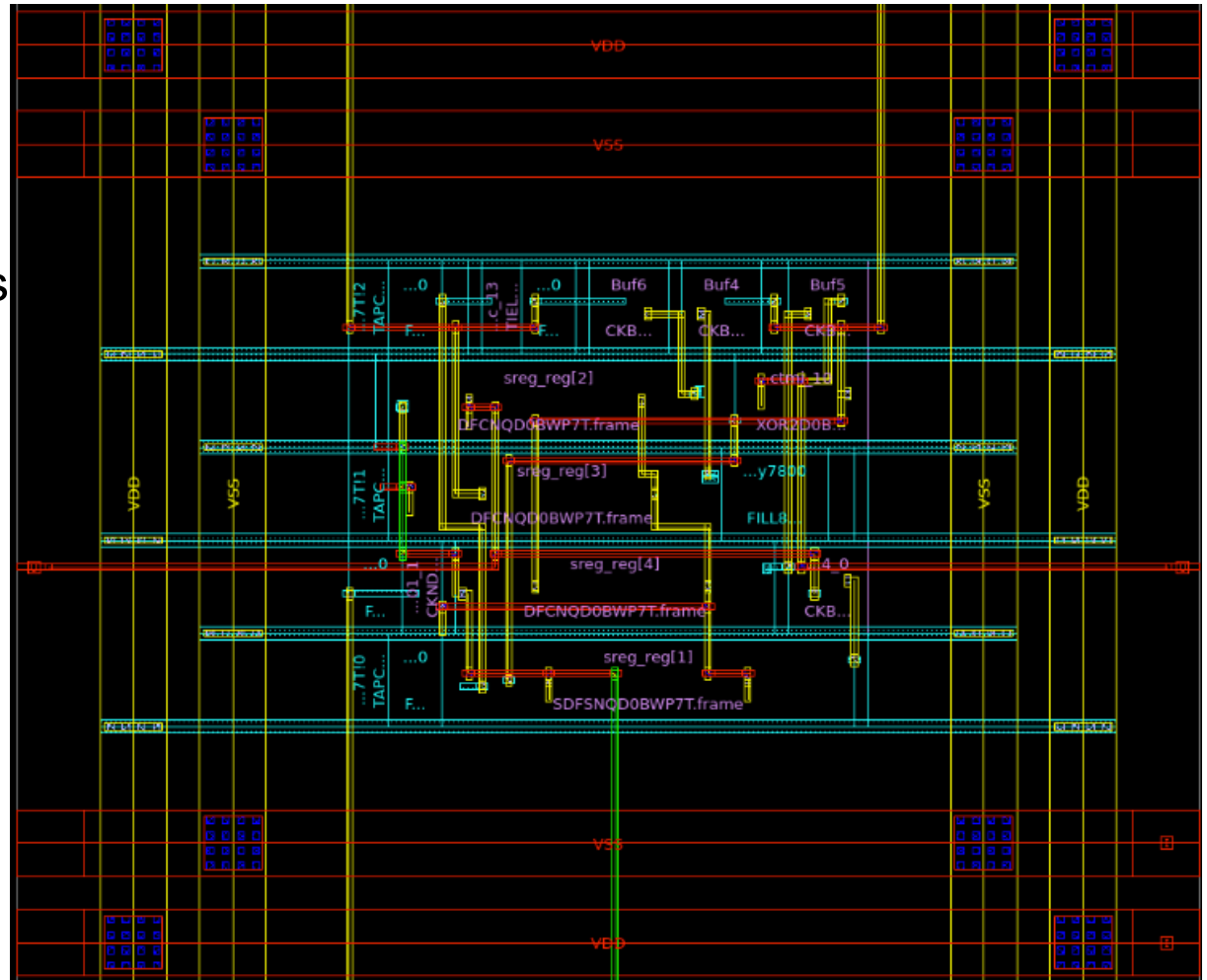
- ◆ PVT stands for Process, Voltage and Temperature
- ◆ Need to tell Place-and-Route tools these variations
- ◆ Ensure circuit works when manufactured
- ◆ Must also work under worst case environment
- ◆ Specified PVT through MMMC file (Tcl format)



Results from PnR in Lab 1 Task 3

◆ To produce physical layout of the LFSR circuit, we need to:

1. Perform floorplanning
2. Perform placement
3. Synthesize the clock tree
4. Perform routing
5. Check the PnR results for errors
6. Export reports and outputs



Simulation using VCS

Testbench (lfsr4_tb.sv)

```
`timescale 1ns/1ps
module lfsr4_tb;
    logic rst;
    logic clk;
    logic [3:0] data_out;

    // instantiate the DUT
    lfsr4 dut (
        .clk(clk),
        .rst(rst),
        .data_out(data_out)
    );

    initial begin
        $display("Start lfsr4 testbench ....");
        clk = 1'b0;
        rst = 1'b1;
        #1 clk = 1'b1;
        #1 clk = 1'b0;
        #1 rst = 1'b0;

        // clock generator
        for (int i=0; i<20; i++) begin
            #1 clk = 1'b1;
            #1 clk = 1'b0;
        end
    end

    initial begin
        $dumpfile("waveform.vcd");
        $dumpvars(1, lfsr4_tb, dut);
    end
endmodule
```

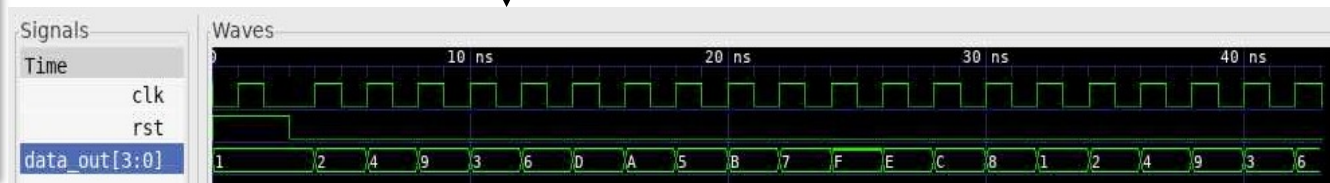
```
## sim-synth: simulate the synthesised netlist, real cells, no wire delay
sim-synth:
    @test "$(FLOW)" = logical || { \
        echo "sim-synth is for FLOW=logical. The fusion flow writes no"; \
        echo "intermediate netlist: placement carries straight through."; \
        exit 1; }
    @test -f $(OUT)/$(DESIGN)_synth.v || \
        { echo "No $(OUT)/$(DESIGN)_synth.v. Run scripts/syn_logical.tcl."; exit 1; }
    $(call run_sim,$(VCS) $$SYN_SIM_MODELS $(OUT)/$(DESIGN)_synth.v $(TB),synth)
```

Makefile

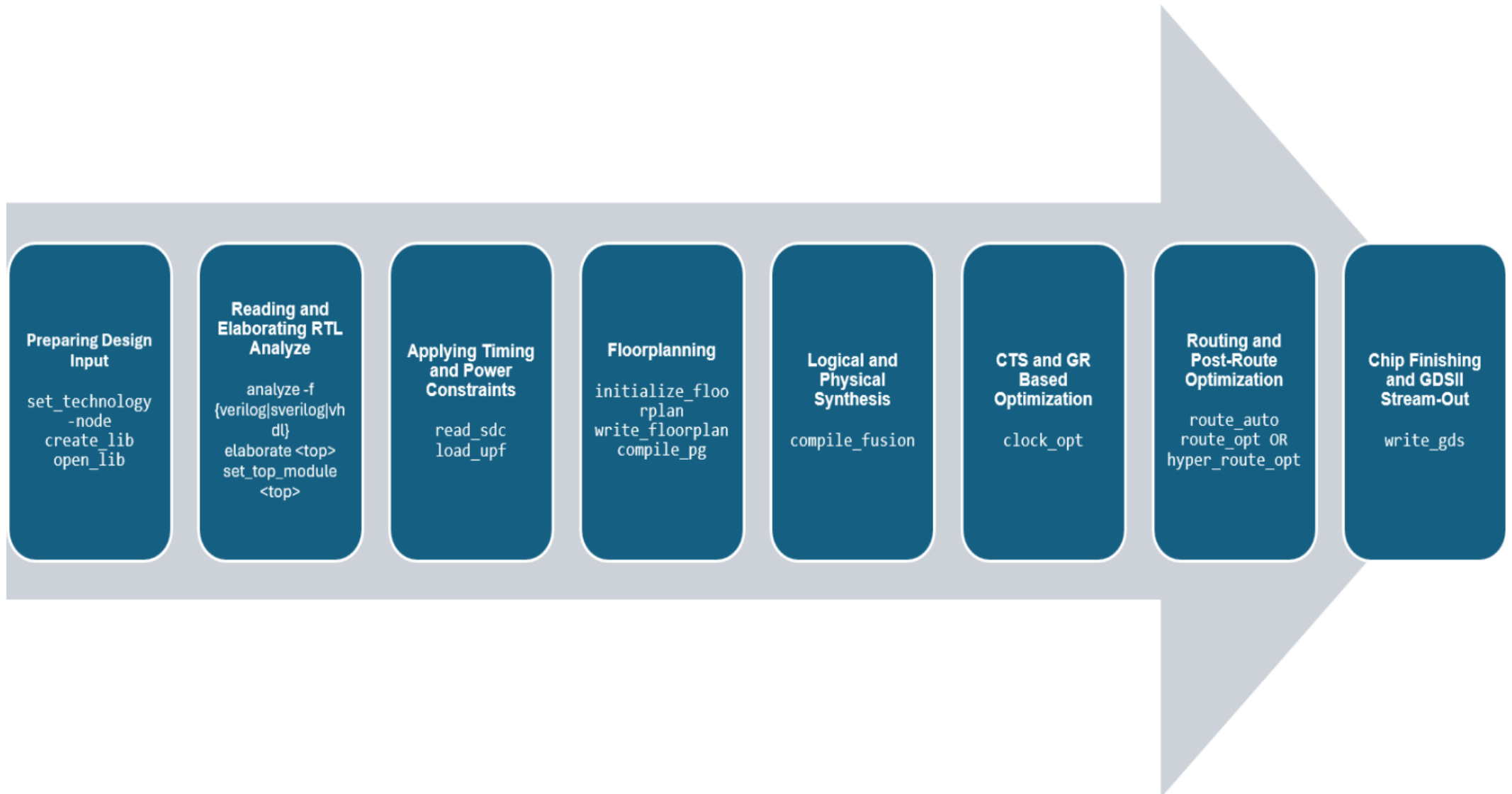
VCS simulator

waveform.vcd

GTKwave



Fusion Compiler's Mega Commands



source: Fusion Compiler User Guide

Further Reading

- ◆ Publicly available standard cell library:

https://www.vlsitechnology.org/html/cells/vsclib013/lib_gif_index.html

- ◆ Synopsys Fusion Compiler User Guide:

https://spdocs.synopsys.com/dow_retrieve/latest/dg/fcolh/pdf/fcug.pdf

- ◆ Fusion Compiler Landing webpage in SolvNet:

https://spdocs.synopsys.com/dow_retrieve/latest/home_public/fc.html

- ◆ Synopsys training page in SolvNet:

<https://training.synopsys.com/learn/course/external/view/elearning/529/fusion-compiler-synthesis-and-design-implementation-jumpstart>

- ◆ “New VLSI Tools”, Gajski&Kuhn ([10.1109/MC.1983.1654264](https://www.coursera.org/learn/new-vlsi-tools))

Access to Documents – SolvNet Registration

SolvNetPlus Account Registration

- Visit SolvNetPlus:
 - <https://solvnetplus.synopsys.com/>
- Click on:
 - "Register – Create Account"
- Fill out the required information
 - University email addresses and Site ID are required
 - Site ID for **Imperial College** is siteID **2727**
 - Registration may take up to 24 hours to process

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Last Name

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Software Community

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Step 1 of 4: Please tell us a bit about yourself. Complete all fields including address.

* First Name

* Last Name

* Job Title

* Email

* Phone

Address

Country

State/Province

City

Zip/Postal Code

Next